AI Job Marketplace Demo Documentation | AI 招聘市场 Demo 文档

English

1. Architecture

This demo is built with Next.js App Router as a full-stack web app.

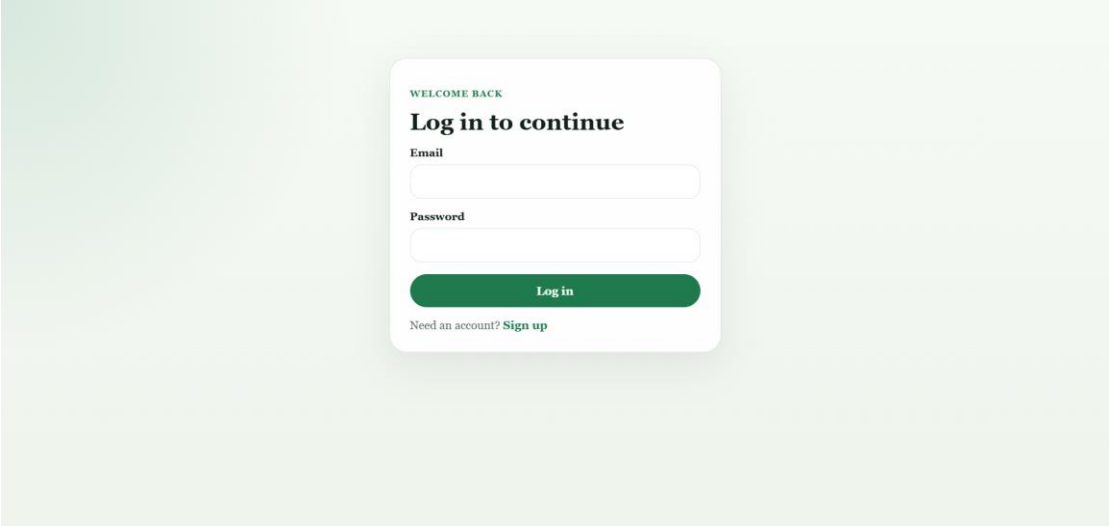
- Frontend: Next.js, React, custom CSS
- Backend: Next.js Route Handlers
- Database/Auth: Supabase
- AI Integration: OpenAI API
- Deployment: Vercel

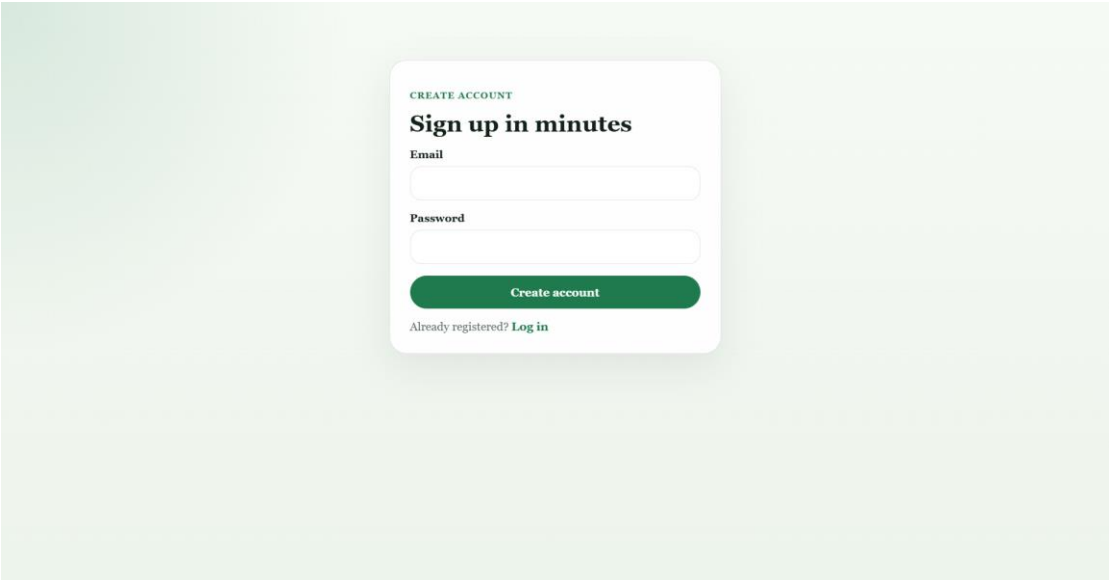
Main modules:

- Authentication: login/signup/session handling
- Dashboard: job matching and job list
- Job Management: create, edit, delete, detail view
- Profile: display name, email, skills
- AI Feature: generate skills from job description

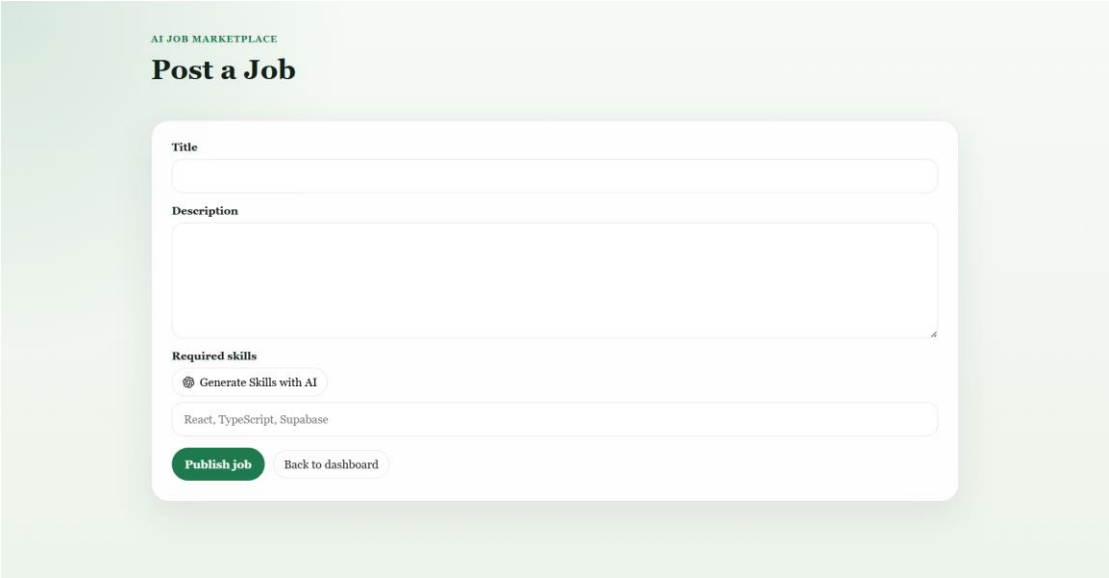
2. Features

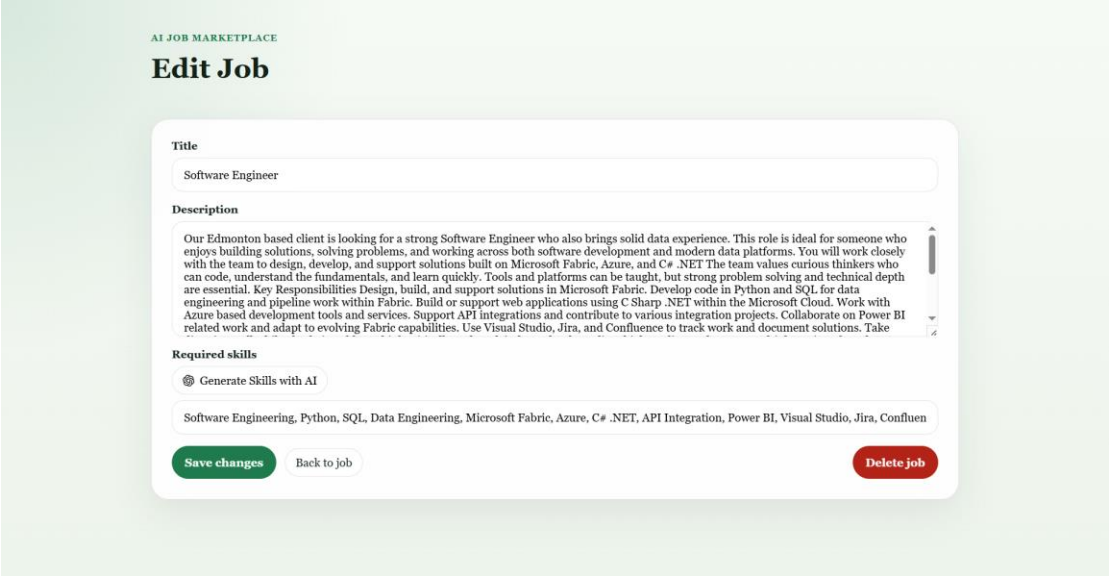
- Email/password authentication



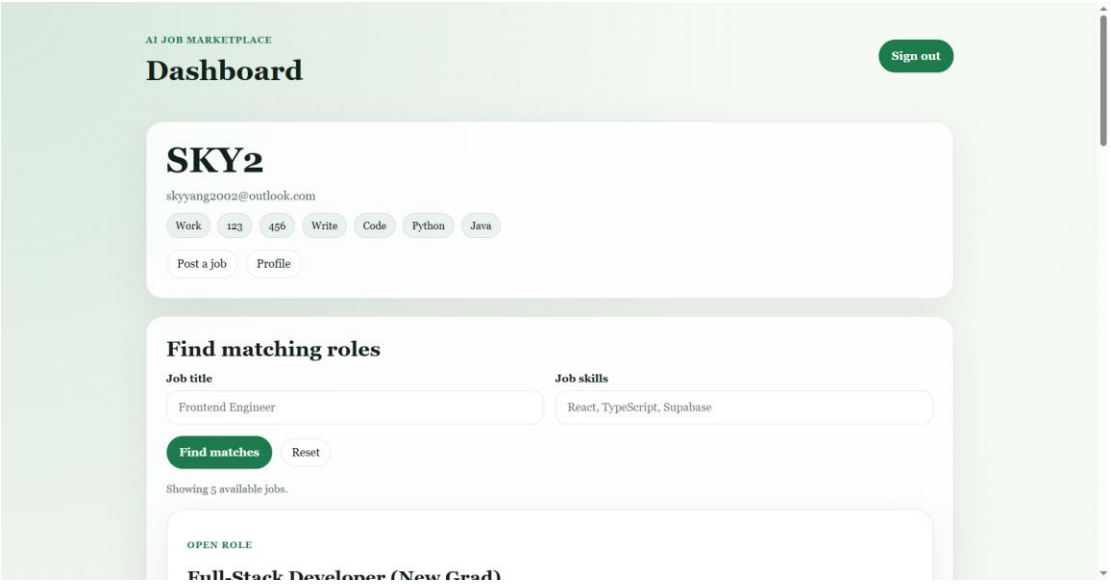


- Employer can create, edit, and delete jobs

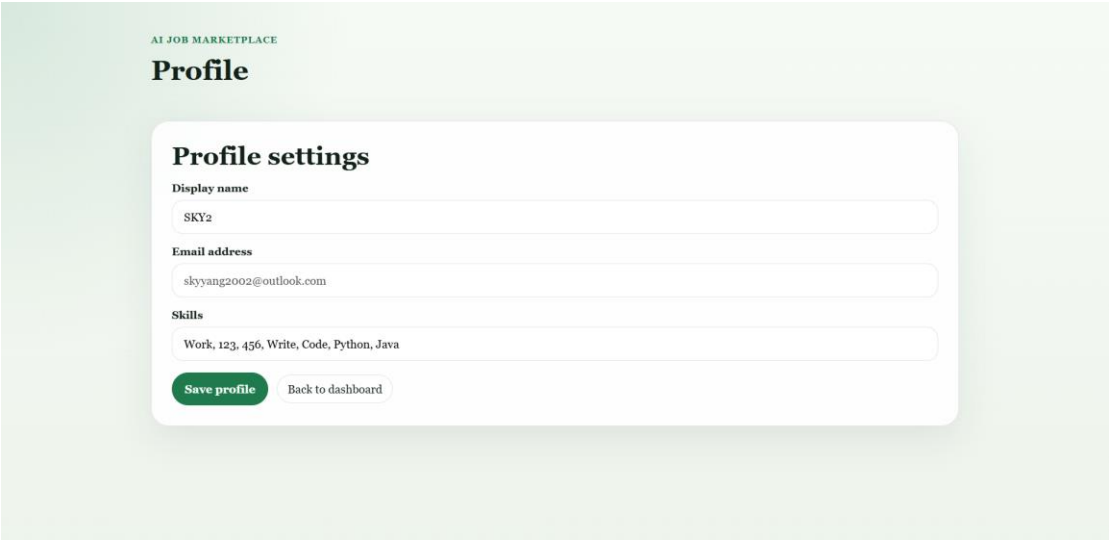




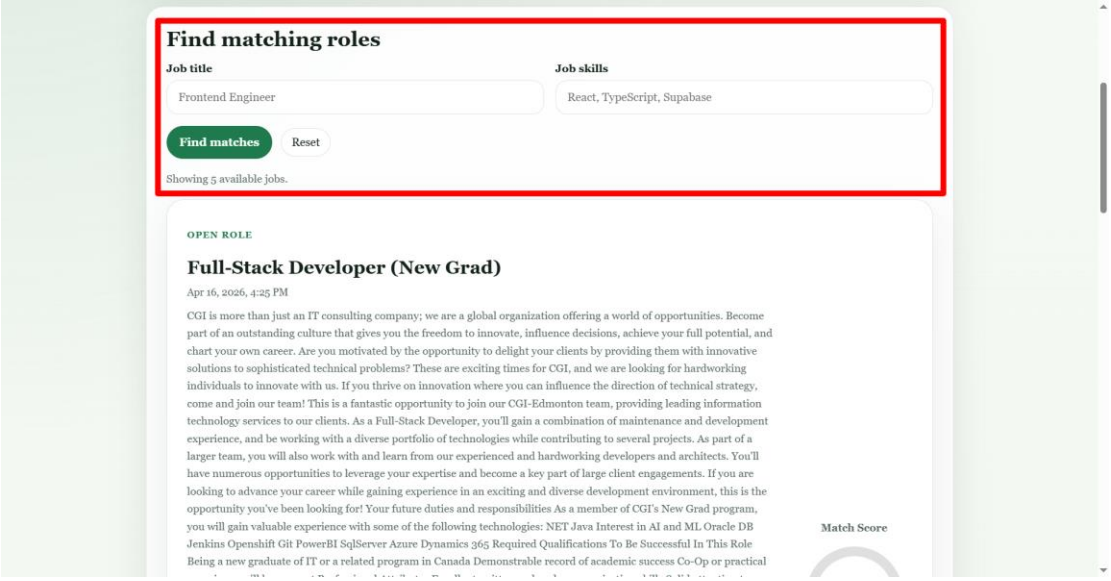
- Jobseeker can browse job cards and job details



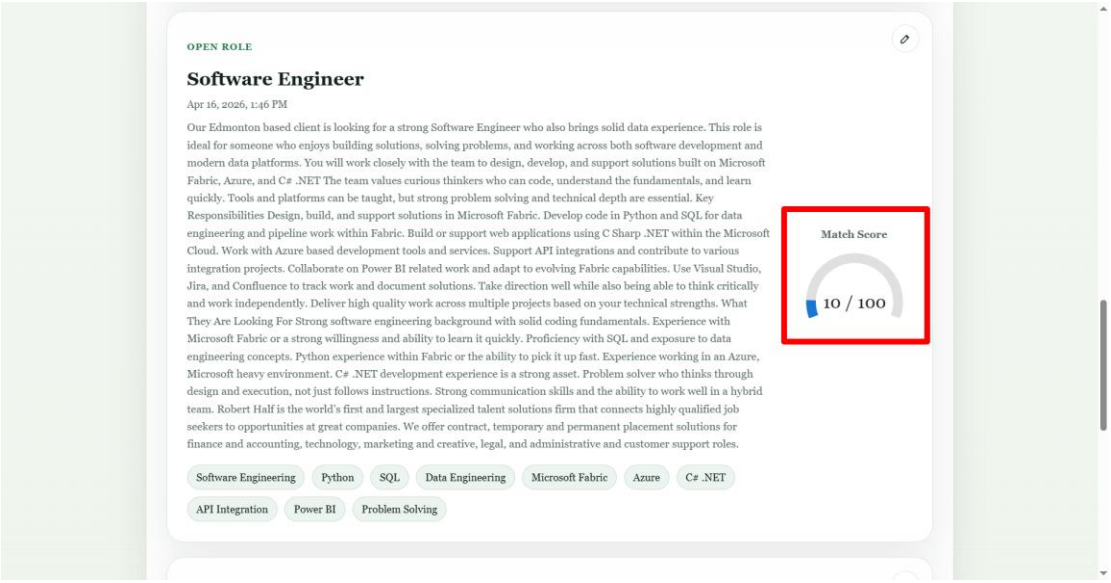
- Profile page with editable display name and skills



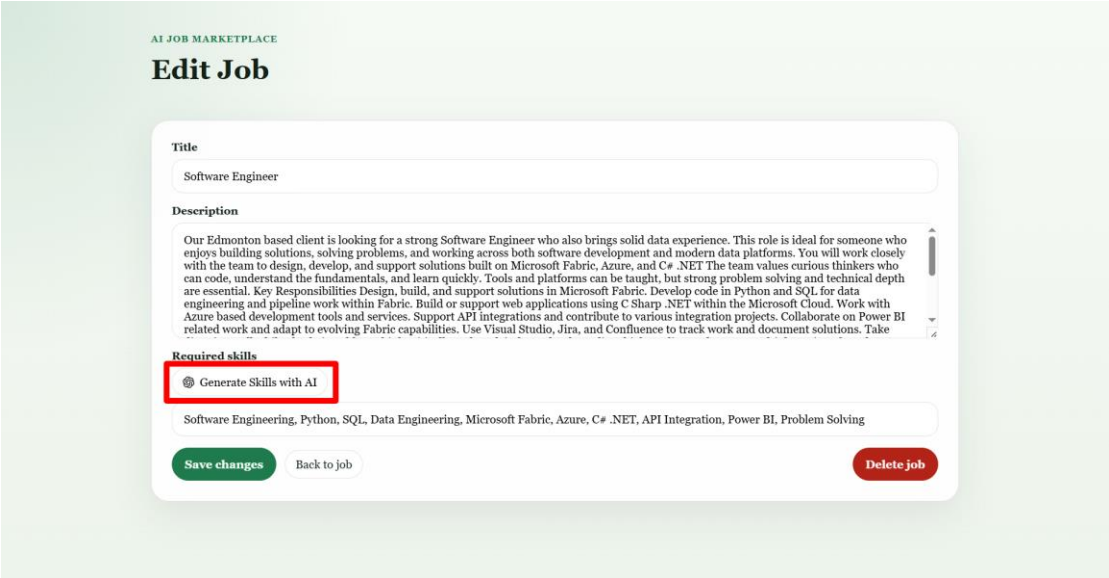
- Server-side job matching



- Match score visualization with gauge chart



- AI-generated skills for job posting/editing



3. AI Usage

AI was used in two stages:

1. ChatGPT
    - Used first to analyze the demo requirements
    - Helped define MVP scope, architecture, and feature breakdown
2. Codex
    - Used to generate and refine the actual code
    - Helped implement authentication, job CRUD(Create, Read, Update, and Delete), profile, AI skill generation, and UI fixes

Key prompts:

- Create a minimal MVP architecture for an AI job marketplace using Next.js App Router. Include authentication, employer job posting, job browsing, matching logic, and one AI feature.
- Generate a simple authentication system using Supabase in Next.js App Router. Requirements: email/password login, signup page, store user session, redirect after login.
- Create a job posting feature in Next.js. Requirements: title, description, required skills, save to Supabase, API route, validation.
- Display match score for each job card on the right side, use MUI Gauge Chart, and use user profile skills to match job required skills only.
- Fix this Vercel build error: useSearchParams() should be wrapped in a suspense boundary at page "/login".
- Clickable area of AI button on edit and post job page is not match the button, fix this problem.

4. Challenges and Solutions

Challenge 1: Vercel deployment required clean lint/build

Problem:

- Vercel enforced lint and build checks
- Small issues like TypeScript warnings, Suspense errors, and unused variables caused

multiple failed deployments
- This required repeated debug and redeploy cycles

Solution:
- Fixed lint/type issues one by one
- Added Suspense boundary for useSearchParams()
- Removed unused variables and implicit any
- Standardized API route typing and auth checks

Prompts that helped:
- Fix these ESLint errors...
- Fix this build error on Vercel...
- Fix this type error: Parameter 'cookiesToSet' implicitly has an 'any' type.

Challenge 2: AI-generated frontend UI was often wrong or misaligned

Problem:
- AI-generated UI sometimes had wrong clickable area, misaligned button positions, and inconsistent card layout
- Some elements looked correct visually but behaved incorrectly
- AI sometimes generated correct logic but weak UI alignment

Solution:
- Refined layout with small targeted prompts instead of regenerating whole pages
- Adjusted CSS step by step
- Separated label/button/input structure when click area was incorrect
- Validated interaction and visual layout together

Prompts that helped:
- Move job-card-score out of job-card-header and put it into job-card-link panel-inner stack
- Tweak the CSS to make the score in the similar position, right middle of the job card
- Clickable area of AI button on edit and post job page is not matching the button, fix this problem

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中文

1. 项目架构

这个 Demo 基于 Next.js App Router 构建，是一个全栈 Web 应用。

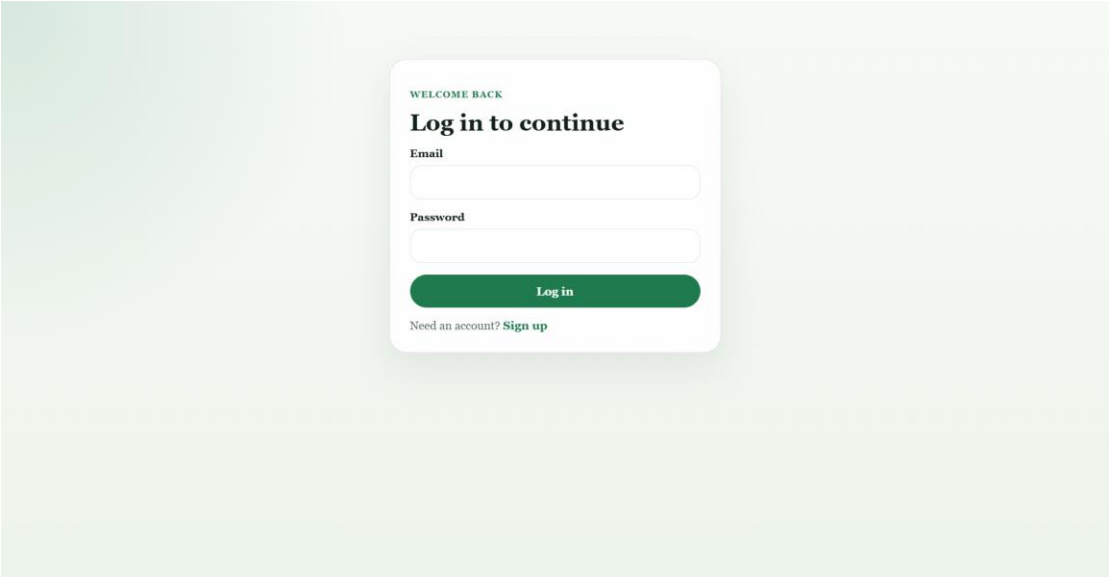
- 前端：Next.js、React、自定义 CSS
- 后端：Next.js Route Handlers
- 数据库/认证：Supabase
- AI 集成：OpenAI API
- 部署：Vercel

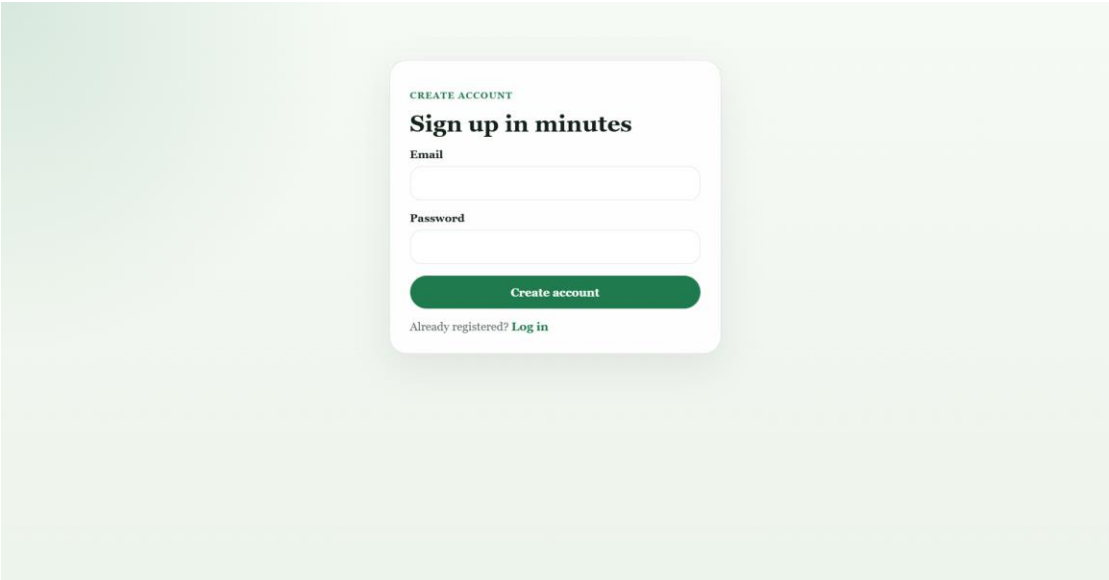
主要模块：

- 用户认证：登录、注册、会话管理
- 控制台：职位匹配、职位列表
- 职位管理：创建、编辑、删除、详情页
- 个人资料：姓名、邮箱、技能
- AI 功能：根据职位描述生成技能

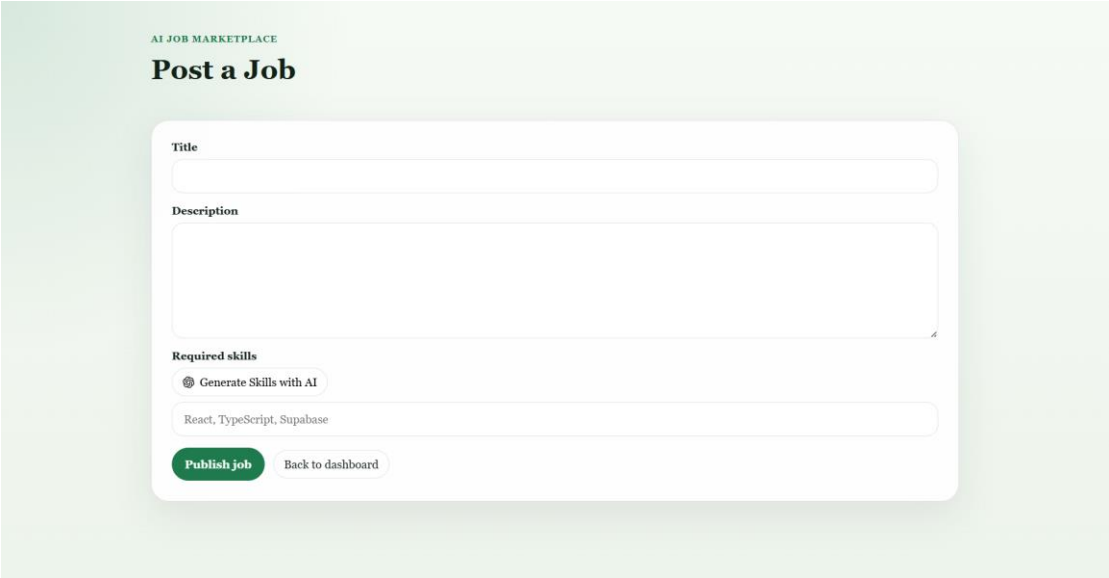
2. 功能

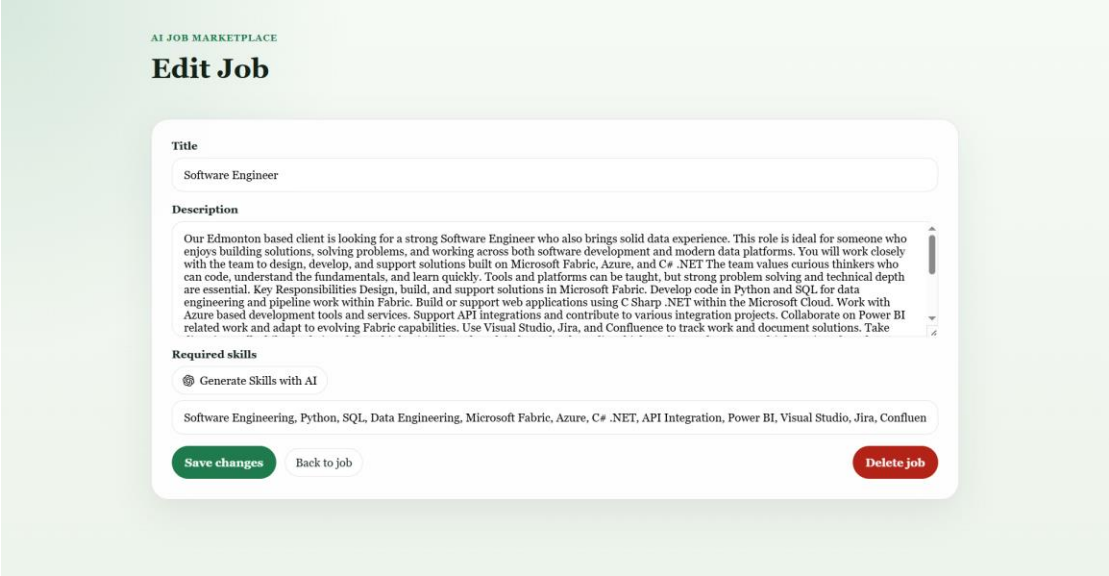
- 邮箱/密码登录注册



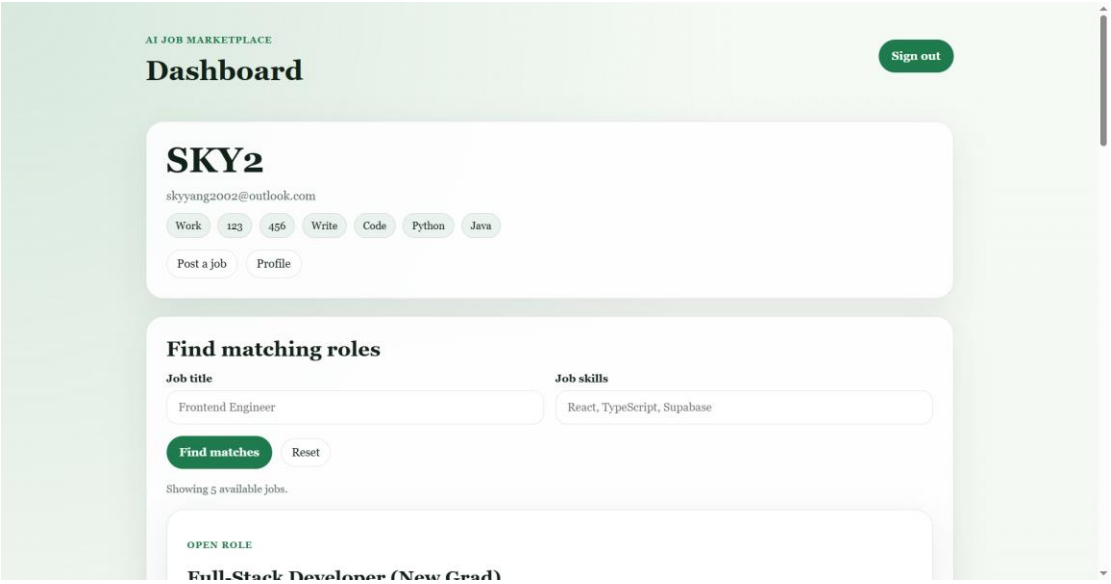


- 雇主可创建、编辑、删除职位

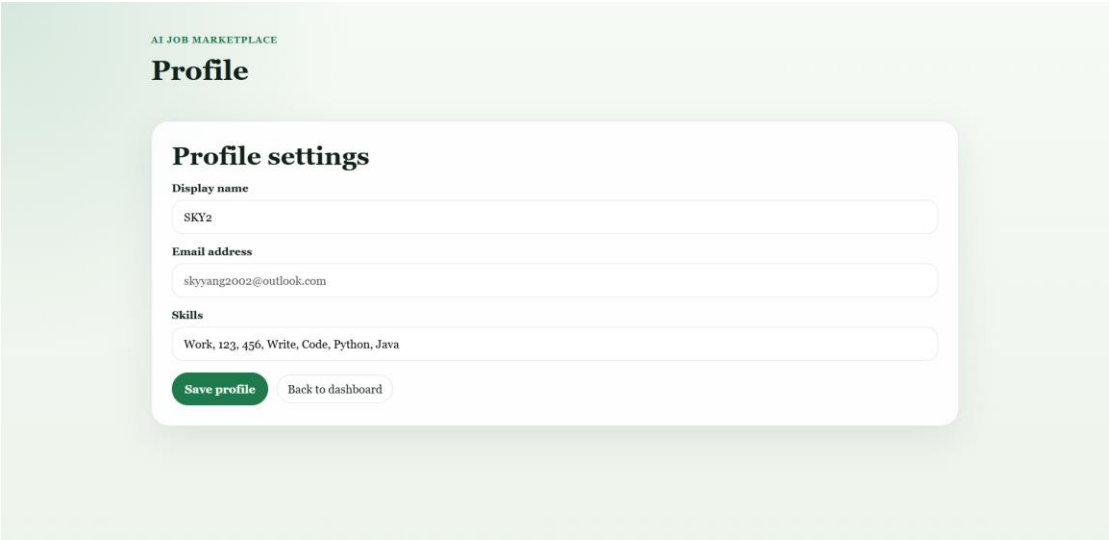




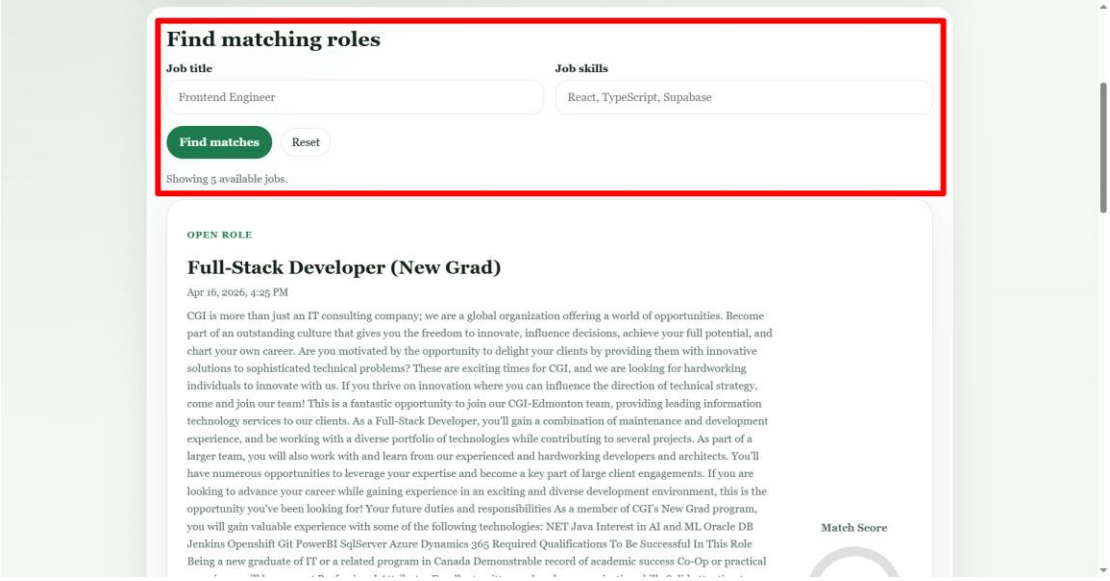
- 求职者可浏览职位卡片和职位详情



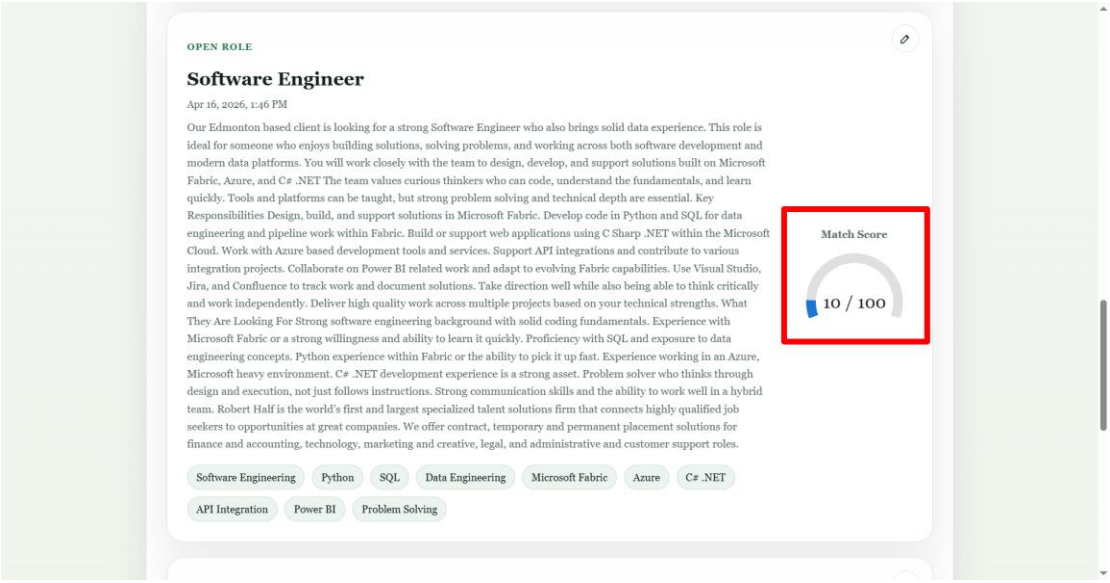
- 个人资料页可编辑显示名称和技能



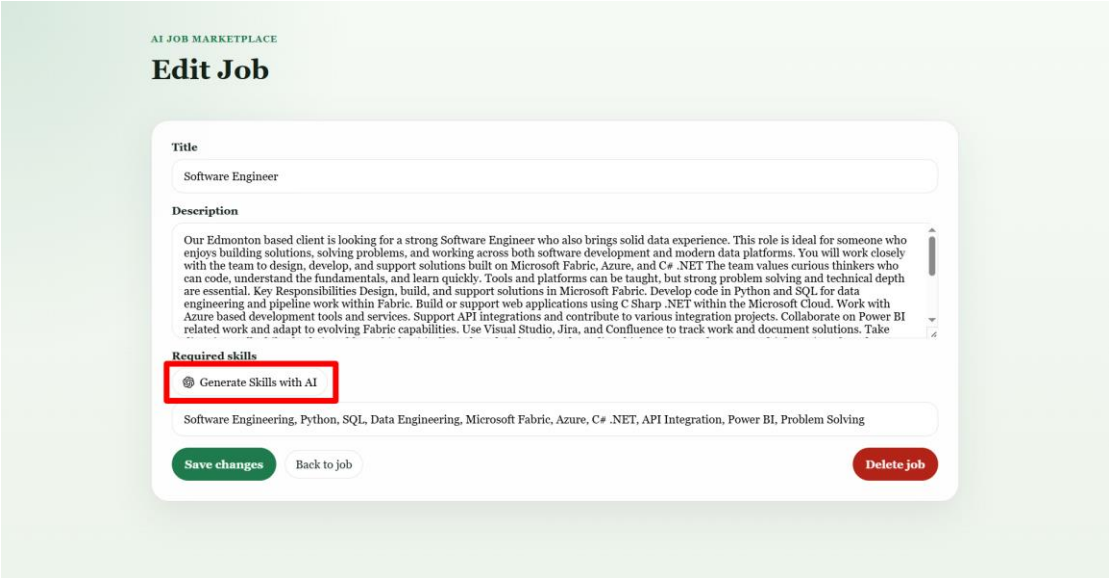
- 服务端职位匹配



- 使用仪表盘显示匹配分数



- AI 自动生成职位技能



3. AI 使用方式

AI 在两个阶段被使用：

1. ChatGPT
    - 用于前期分析 Demo 需求
    - 帮助梳理 MVP 范围、架构和功能拆分
2. Codex
    - 用于实际生成和修改代码
    - 实现了认证、职位增删改查、个人资料、AI 技能生成和 UI 修复

关键提示词：

- Create a minimal MVP architecture for an AI job marketplace using Next.js App Router. Include authentication, employer job posting, job browsing, matching logic, and one AI feature.
- Generate a simple authentication system using Supabase in Next.js App Router. Requirements: email/password login, signup page, store user session, redirect after login.
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- Fix this Vercel build error: useSearchParams() should be wrapped in a suspense boundary at page "/login".
- Clickable area of AI button on edit and post job page is not match the button, fix this problem.

4. 挑战与解决方案

挑战 1：Vercel 部署要求 lint/build 全部通过

问题：

- Vercel 会强制检查 lint 和 build
- 一些小问题，如 TypeScript 报错、Suspense 错误、未使用变量，都会导致部署失败

- 因此需要多次调试和重新部署

解决方案：

- 逐个修复 lint/type 错误
- 为 useSearchParams() 增加 Suspense
- 清理未使用变量和隐式 any
- 统一 API 路由和认证检查

使用的提示词：

- Fix these ESLint errors...
- Fix this build error on Vercel...
- Fix this type error: Parameter 'cookiesToSet' implicitly has an 'any' type.

挑战 2：AI 生成的前端 UI 经常出现错位或交互不准确

问题：

- AI 生成的 UI 有时会出现点击区域不对、按钮位置错位、卡片布局不一致
- 有些界面看起来正确，但交互行为不对
- 有些代码逻辑虽然正确，但界面对齐效果不好

解决方案：

- 不一次性重写，而是通过小步迭代优化
- 优先微调 CSS，而不是整块重生成
- 当点击区域异常时，拆分 label/button/input 结构
- 同时检查视觉效果和交互行为

使用的提示词：

- Move job-card-score out of job-card-header and put it into job-card-link panel-inner stack
- Tweak the CSS to make the score in the similar position, right middle of the job card
- Clickable area of AI button on edit and post job page is not matching the button, fix this problem